

Vision-Guided Quadrotor Drone – Object Tracking

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Introduction

- Overall goal: Use drone to capture images of a building's exterior to use for 3D mapping
- My task: Enable drone to autonomously track an unmanned ground vehicle (UGV)
- Fields: computer vision, machine learning, robotics, and autonomous systems
- Drones used: Qualcomm Flight RB5 from ModalAI, Tello EDU from Ryze Robotics (for testing).
- UGV used: Jackal from Clearpath Robotics.



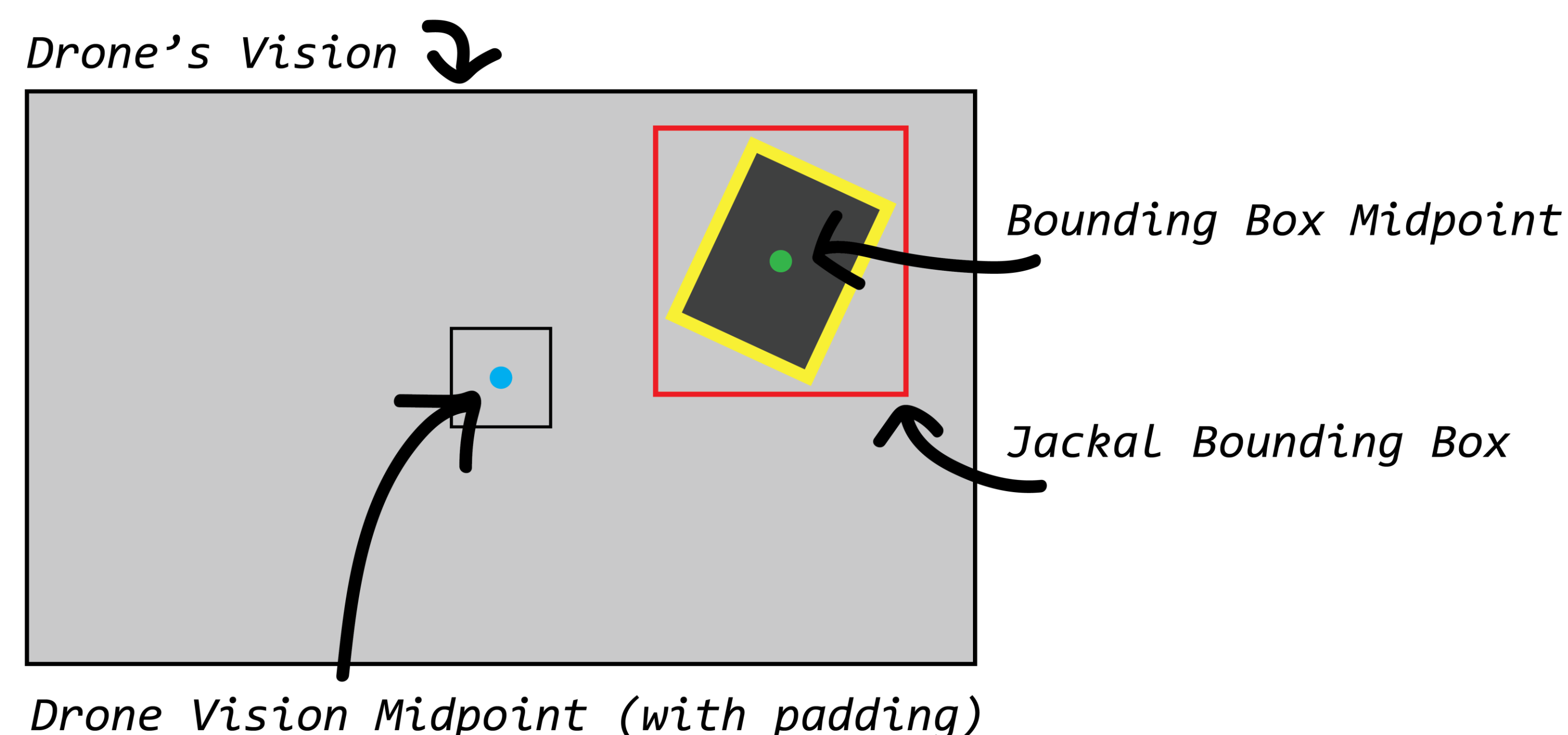
Object Detection

- Gathered and annotated images of Jackal
- Trained an object recognition model, YOLOv5s
- Tested trained model's performance to find limitations and ways to improve
- Repeated process while increasing dataset with images taken in lab, Google images, photoshopped images with different backgrounds/varying degrees of occlusion, and images from significant height (50+ feet high)



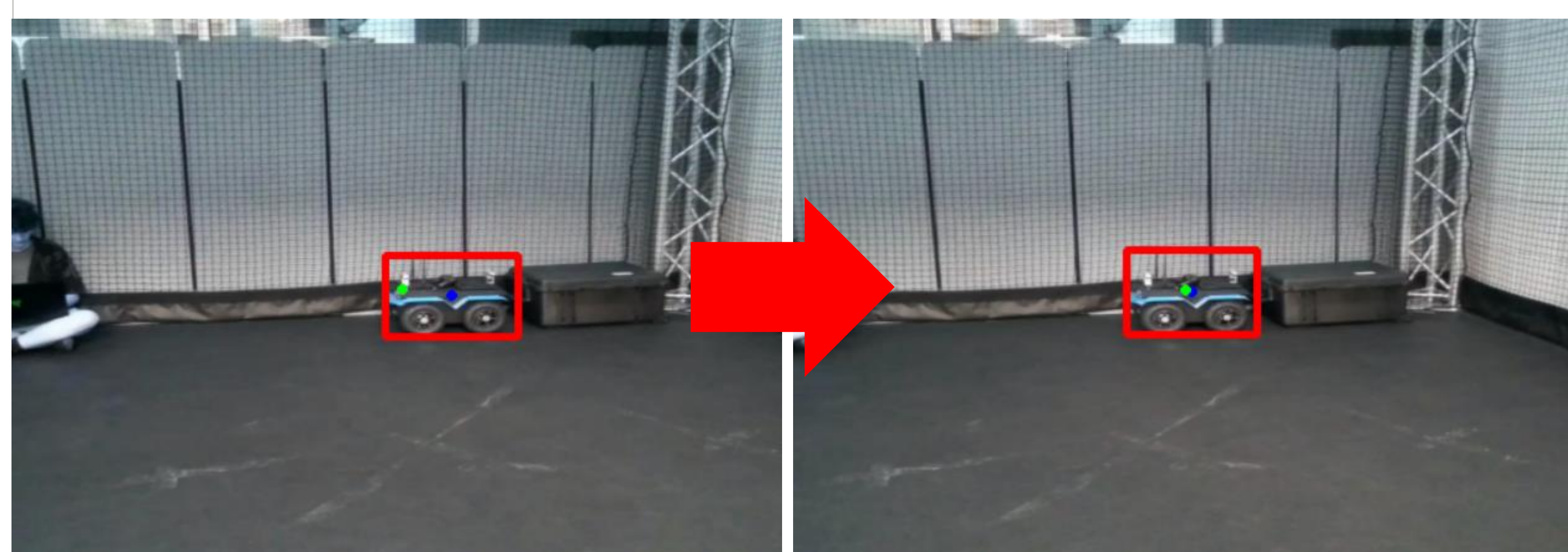
Object Tracking

- Worked out method for creating flight commands from bounding box drawn around detected Jackal



If bounding box midpoint is not close to the center of the drone's vision, move drone forward/backward, left/right to align midpoints. In this instance, drone would move right and forward to align itself over the detected Jackal. (Method will be improved and tweaked as testing reveals limitations.)

- Tested tracking method on Tello drone (safer for testing than RB5 drone)



Conclusion

- Object recognition model can accurately identify the Jackal with high confidence in a variety of environments, at a variety of heights (50+ feet), with varying degrees of occlusion



Can add images of everyday objects to dataset and retrain to prevent incorrect identifications

- Initial tests of tracking method work but need additional testing and improvements
 - What should prompt the drone to turn?
 - What if the drone loses sight of the Jackal?
 - How much "padding" should there be around midpoint differences?